Johannes Roth

Machine Learning Engineer

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Available from January 2027

Machine learning engineer with experience taking ML and data projects from initial requirements to deployment and operation. Background in production computer vision, large-scale data processing, and independent research in computational neuroscience.

EXPERIENCE

PhD Researcher - Computational Cognitive Neuroscience May 2022 – Present
Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen

- Conducted independent research at the intersection of **machine learning and computational neuroscience**, focusing on vision models and their relationship to human brain activity.
- Developed methods for model evaluation and experimental design, combining computational analyses with fMRI studies.
- Developed **GPU-based ML pipelines** for dataset curation and feature extraction, and released large-scale image datasets for computer-vision research.
- Independently developed and maintained **cloud services for sharing research datasets**, public websites, and tools for organizing and annotating data.
- Supervised working students and interns and collaborated with researchers across computer science and neuroscience.

Research Scientist - Machine Learning in Medicine Jun 2021 – May 2022
ScaDS.AI Dresden/Leipzig

- Developed and evaluated deep-learning models for **medical image analysis**, including brain-tumor segmentation.
- Built predictive models for clinical data, with a focus on mortality prediction and uncertainty estimation.

Data Scientist / ML Engineer (Working Student) Nov 2019 – May 2021
CHECK24, Travel Vertical

- Independently designed and deployed a **production computer-vision service** for **20 million hotel images**, covering model development and backend engineering.
- Prototyped additional computer-vision applications for image retrieval, duplicate detection, and image-quality assessment.
- Optimized recommendation-data pipelines through incremental processing, reducing update time by over **95%**.

Full-Stack Developer (Freelance) Oct 2020 – May 2021
Kimetric UG

- Designed and delivered **Django-based websites** to present research, publications, and public datasets.
- Gathered client requirements and managed deployment, Linux hosting, and ongoing application maintenance.

Data Scientist (Working Student) Oct 2018 – Sep 2019
Webdata Solutions GmbH (now Vistex)

- Developed machine-learning models for **product matching and image classification** in e-commerce.
- Improved image-based matching accuracy from **12% to 83%** against a classification baseline in an internal cross-retailer evaluation.

Data Analyst (Working Student) Feb 2018 – Aug 2018
Mercateo Services GmbH & Co. KG

- Developed data-integration, reporting, and testing workflows for an **AWS-based data warehouse** using Apache NiFi and Pentaho.

TECHNICAL SKILLS

- **ML & Analysis:** Python, SQL, PyTorch, TensorFlow, scikit-learn; computer vision, embeddings, model evaluation, statistical analysis, experimental design.
- **Software & Data:** FastAPI, Flask, Django, Redis, Docker, Git, CI/CD; data pipelines, similarity search, dataset curation.
- **Cloud & Compute:** AWS (S3, IAM), Linux, SLURM/HPC, multi-GPU processing.
- **Languages:** German (native), English (fluent).

OPEN SOURCE

- Contributed to **thingsvision**, an open-source library for computer-vision research, with work on software architecture, model integration, and memory efficiency.

EDUCATION

Doctoral Candidate (Dr. rer. nat.) May 2022 – Present
University of Giessen
Research focus: machine learning and computational neuroscience.

M.Sc. Computer Science 2017 – 2021
Leipzig University; grade 1.2

- **Master's thesis (grade 1.1):** Applied generative deep learning to study visual processing in the human brain.

B.Sc. Business Information Systems 2014 – 2017
Leipzig University; grade 1.5

SELECTED PUBLICATIONS

- F. P. Mahner*, **J. Roth***, et al. *Characterizing Universal Object Representations Across Vision Models*. **NeurIPS 2026** (accepted). Shared first author.
- **J. Roth***, Y. Duan*, et al. *Ten principles for reliable, efficient, and adaptable coding in psychology and cognitive neuroscience*. Communications Psychology, 2025. Shared first author.
- **J. Roth** et al. *Multi-plane UNet++ Ensemble for Glioblastoma Segmentation*. MICCAI BrainLes 2021 workshop proceedings, published 2022.

RECOGNITION & SERVICE

- Presented research at **CCN 2025 (oral)**.
- Elected representative for **180+ doctoral researchers** at MPI CBS (2023–2024).
- **CMBB Replication Award (2025)** for contributions to reliable coding practices in neuroscience.